Nematollah (Emad) Saeidi

AI Engineer | OeAD Grant @ DPU (Austria) | PhD | Medical Image Analysis | Computational Pathology | GNNs & Foundation Models | RAG, Hybrid Search & Document AI

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Summary

- AI engineer and researcher (PhD in Artificial Intelligence) with 8 years of experience across applied machine learning, computer vision, generative AI, information retrieval, and medical image analysis. Currently the sole AI specialist at a startup, responsible for multilingual hybrid search and RAG over enterprise data, and for intelligent document processing with OCR, vision-language models, and LLM-based structured extraction. Research focuses on graph neural networks and foundation models for medical image retrieval and computational pathology, published in Q1/Q2 journals.

Education

PhD **University of Isfahan**, Artificial Intelligence — Isfahan, Iran
Thesis: Content-based Medical Image Retrieval Using Graph Neural Networks. — Jan 2019 – Oct 2026
- Thesis: Content-based Medical Image Retrieval Using Graph Neural Networks.

MSc **University of Isfahan**, Artificial Intelligence — Isfahan, Iran
Thesis: Learning Multi-Objective Binary Features for Image Representation and Recognition. — Jan 2015 – Jan 2017
- Thesis: Learning Multi-Objective Binary Features for Image Representation and Recognition.

BSc **Islamic Azad University Najaf Abad**, Software Engineering — Isfahan, Iran
Thesis: Intelligent Agent-based Social Simulation Model using Reinforcement Learning. — Jan 2010 – Jan 2013
- Thesis: Intelligent Agent-based Social Simulation Model using Reinforcement Learning.

Experience

Startup, AI Engineer — Tehran, Iran (Remote)
Sole AI specialist at a startup, with end-to-end responsibility for the company's AI services. — Apr 2026 – present — 8 months
- Multilingual (Persian/English) hybrid search over enterprise data: lexical and semantic retrieval, reranking, LLM-based relevance scoring, and grounded answer generation.
- Asynchronous OCR and vision-language-model pipelines that convert documents and conversations into structured data, with LLM-based extraction, validation, and template-driven conversational generation.
- Multi-provider LLM orchestration with failover and local fallback, PII redaction, observability, streaming APIs, and fully on-premises deployment for sensitive data.

Hyrcan Technology Development Co., AI Engineer — Tehran, Iran (Hybrid)
Develop local generative AI and real-time identity recognition applications. — Nov 2025 – Mar 2026 — 5 months

- Built a local RAG-based voice assistant with LangChain, Ollama, JSON/ChromaDB, and Sentence Transformers.
- Applied few-shot prompting and conversation memory to support contextual voice dialogue.
- Integrated Whisper speech-to-text, RAG-enhanced LLM generation, and Coqui text-to-speech into a complete speech pipeline.
- Implemented real-time speaker recognition with NVIDIA NeMo TitaNet and Resemblyzer, incorporating voice activity detection and acoustic feature extraction.
- Implemented real-time face recognition pipelines using YOLO for detection and ArcFace ResNet100 / InsightFace for recognition.

Danube Private University Austria (DPU), MIAAI Group, AI Researcher — TFZ, Austria (Hybrid)
Research medical image retrieval and histopathology image analysis in the Medical Image Analysis & Artificial Intelligence (MIAAI) group. — Sept 2023 – present — 3 years 3 months

- Investigate graph neural networks and foundation models for content-based medical image retrieval.
- Work on histopathological image classification using nuclei-based graph convolutional networks and foundation-model features.
- Apply image classification, nuclei segmentation, and generative modeling methods to medical image analysis.

FANAP Co., Marketing Data Scientist — Tehran, Iran (Remote)
Supported AI service evaluation and business development while contributing to applied research projects. — July 2021 – Sept 2023 — 2 years 3 months

- Collaborated with cross-functional teams to troubleshoot and optimize more than 10 deployed AI services.
- Designed and executed evaluations of model accuracy, speed, and reliability; identified and reported anomalies in AI systems.
- Drafted requests for proposals (RFPs) and business plans for initiatives in the AI open innovation ecosystem.
- Authored technical articles and research briefs for the FANAP Research Team.
- Medical image search engine (Dec 2022–Sep 2023): worked on retrieval and classification using attention-based adversarial variational graph autoencoders, pretrained models, and approximate nearest-neighbor search.
- Drawing-robot motion planning (Mar–Jul 2022): developed image-processing methods to extract contours for a drawing robot.
- Worked within an agile process using daily standups, sprint retrospectives, backlog grooming, and continuous integration in two-week sprints.

FANAP Co., Data Scientist — Tehran, Iran (Hybrid)
Developed computer vision, recommendation, and customer analytics applications at FANAP. — Oct 2018 – July 2021 — 2 years 10 months

- Object detection API (Sep 2020–Jul 2021): implemented detection and segmentation workflows using few-shot models and transfer learning.
- Video surveillance attendance API (Jan–Mar 2021): implemented an attendance and registration application.
- Automatic number plate recognition (Jan 2021): implemented vehicle registration, vehicle-data search, and ranked record display by license plate.
- Visual search API (Jan–Sep 2020): combined object detection and recognition, color extraction, image crawling, transfer learning, and approximate nearest-neighbor search.
- Customer analytics and clustering library (Aug 2019–Jan 2020): developed SaaS metrics and KPIs, segmentation, and cluster ensembles for numeric and categorical grocery-retail data.
- Transaction analysis (Aug 2019–Jan 2020): clustered customer behavior in credit-card transactions to support fraud detection, including money-laundering and tax-evasion analysis.
- Recommendation system (Mar–Aug 2019): developed sequence-to-sequence ranking, association rules, hybrid/content-based/collaborative filtering, customer clustering, and feature engineering for candidate generation.
- Marketing analysis and product prediction (Oct 2018–Feb 2019): applied CatBoost, XGBoost, LightGBM, ensemble methods, and multilayer perceptrons across business datasets.
- Collaborated on fraud detection, chatbot, video surveillance, recommendation, and customer behavior analysis services.
- Worked in two-week agile sprints with daily standups, retrospectives, backlog grooming, and continuous integration.

Publications

Unfolding Graph Convolutional Networks: Architectures, Use Cases, and Trends — Jan 2026
Nematollah Saeidi, Vahid Hajipour, S. Hekmat, B. Shoushtarian
10.1016/j.asoc.2026.115122 (Applied Soft Computing (IF~8.2, Q1))

Leveraging Medical Foundation Model Features in Graph Neural Network-Based Retrieval of Breast Histopathology Images — Jan 2026
Nematollah Saeidi, H. Karshenas, B. Shoushtarian, S. Hatamikia, R. Woitek, Amirreza Mahbod
10.1002/ima.70346 (International Journal of Imaging Systems and Technology (IF~3.3, Q2))

Evaluating Pre-trained Convolutional Neural Networks and Foundation Models as Feature Extractors for Content-based Medical Image Retrieval — Jan 2025
Amirreza Mahbod, Nematollah Saeidi, S. Hatamikia, R. Woitek
10.1016/j.engappai.2025.110571 (Engineering Applications of Artificial Intelligence (IF~8, Q1))

Integrating Foundation Model Features into Graph Neural Network and Fusing Predictions with Standard Fine-Tuned Models for Histology Image Classification — Jan 2025
Nematollah Saeidi, N. Torbati, R. Woitek, Amirreza Mahbod
10.3390/bioengineering12121332 (Bioengineering, 12, 1332 (IF~3.7, Q2))

Incorporating Shape Dependent Power Law in Motion Planning for Drawing Robots — Jan 2024
M. Abedinzadeh Shahri, Nematollah Saeidi, Vahid Hajipour
10.1016/j.robot.2024.104801 (Robotics and Autonomous Systems, 182, 104801 (IF~4.3, Q1))

Artificial Intelligence and Clinical Decision Making: Approaches and Challenges Jan 2022
Nematollah Saeidi, M. Torabi
10.22034/JAISIS.2022.371239.1054 (Journal of Applied Intelligent Systems and Information Sciences, 3(2), 43-55)

Attention-Based and Positional-Aware Neural Networks for Next-Item Recommendation Jan 2021
Nematollah Saeidi, H. Shahamat
10.22034/JAISIS.2021.304225.1033 (Journal of Applied Intelligent Systems and Information Sciences, 2(1), 29-40)

Single Sample Face Recognition Using Multicross Pattern and Learning Discriminative Binary Features Jan 2019
Nematollah Saeidi, H. Karshenas, H. M. Mohammadi
10.1080/19361610.2019.1581877 (Journal of Applied Security Research, 14(2), 169-190 (IF~1.2, Q2))

Learning Multi-Objective Binary Features for Image Representation Jan 2017
Nematollah Saeidi, H. Karshenas, H. M. Mohammadi
10.1109/ICCKE.2017.8167927 (2017 7th International Conference on Computer and Knowledge Engineering (ICCKE), Mashhad, pp. 47-52, IEEE)

Projects

Content-Based Medical Image Retrieval (CBMIR) Danube Private University Austria (DPU) Jan 2025
Explores vision and vision-language foundation models for content-based medical image retrieval across 2D and 3D medical images, and investigates the impact of image size on retrieval performance.

Medical Image Retrieval with Foundation Models Jan 2025
Evaluation of pre-trained CNNs and medical foundation models as feature extractors for content-based medical image retrieval.

Histology Image Classification with GNNs Jan 2025
Medical image classification using foundation model features in a graph neural network.

Contour Drawing for a Robot Motion Planning System FANAP Jan 2024
Python code to generate contours from input images based on shape-dependent characteristics, as input for a robot motion planning system implemented in MATLAB.

Automatic Number Plate Recognition System FANAP Jan 2021
Vehicle registration, vehicle data search, top records by license plate, and attendance management for tracking user activity within specified time and date ranges.

Development of a ChatBot UI FANAP Jan 2021
Developed a chatbot user interface.

Fraud Detection FANAP Jan 2021
Web-based system for detecting potential fraudulent activities from user transaction data, combining data processing and user behavior analysis to identify anomalies or fraud risks.

Object Detection App FANAP Jan 2020
Flask-based web application for detecting objects in images and retrieving visually similar images using deep learning and computer vision techniques.

Video Surveillance-Based Attendance and Register App — FANAP, Jan 2021
System for managing and tracking employee attendance, work hours, and personal data, with biometric authentication, data registration, and reporting.

Visual Search App — FANAP, Jan 2020
Visual search combining object detection, recognition, color extraction, and background subtraction, with web crawling, transfer learning, and approximate nearest-neighbor search.

Customer Behavior Analysis — FANAP, Jan 2019
Clustering of customer behavior patterns in grocery retail using SaaS metrics, KPIs, segmentation, cluster ensembles, and numeric and categorical clustering methods.

Recommendation System — FANAP, Jan 2019
Sequence-to-sequence ranking, association rules, hybrid/content-based/collaborative filtering, customer clustering, feature engineering for candidate generation, and data cleansing and database management.

Marketing Analysis and Product Prediction — FANAP, Jan 2018
Marketing analysis and product prediction using CatBoost, XGBoost, LightGBM, ensemble methods, and multilayer perceptrons across business datasets.

Learning Multi-Objective Binary Features for Image Representation and Recognition — University of Isfahan, Jan 2016
MATLAB implementations for extracting robust binary features using Multi-Objective Binary Feature (MOBF) for face recognition and image representation.

Standard Sugarscape Model using Reinforcement Learning — Islamic Azad University, Najafabad Branch, Jan 2012
Agent-based social simulation model using reinforcement learning.

Awards

Ernst Mach Grant, OeAD (Austrian Agency for International Cooperation in Education and Research) — Jan 2024
Research stay at Medical Image Analysis & Artificial Intelligence (MIAAI), Danube Private University (DPU), Austria.
OeAD

Ernst Mach Grant, OeAD (Austrian Agency for International Cooperation in Education and Research) — Jan 2023
Research stay at Medical Image Analysis & Artificial Intelligence (MIAAI), Danube Private University (DPU), Austria.
OeAD

Third place in the call to solve business challenges — Jan 2021
FANAP Co.

Exceptional Talent Award for the Ph.D. Program in Artificial Intelligence (Without National Entrance Exam) — Jan 2019
University of Isfahan

Second in the graduating class, MSc in Computer Engineering (Artificial Intelligence) — Jan 2017
University of Isfahan

Top two percent of graduated students, ranked 3rd among 158 BSc graduates in Computer Engineering — Jan 2013
Islamic Azad University Najaf Abad

Top Student, scholarships for two semesters Jan 2011
Islamic Azad University Najaf Abad

Skills

Generative AI and LLMs
Generative AI, GenAI models, Large Language Models (LLM), Local LLMs, Retrieval-Augmented Generation (RAG), Prompt Engineering, Fine Tuning, Finetuning Large Language Models, Pretraining LLMs, Hugging Face, Transformers, Vector indexing, Voice Chat, LangChain, Ollama, Sentence Transformers, Whisper, Coqui TTS, ChromaDB

Medical Image Analysis
Medical Image Analysis, Histopathology image analysis, Digital Pathology, Foundation Models, Graph Neural Networks, Image Retrieval, Image classification

Computer Vision
Computer Vision, OpenCV, Detectron2, Digital Image Processing, Image Analysis, Image Search, Facial Recognition, Speaker Recognition, Transfer Learning, Approximate Nearest Neighbor Search, ImageAI, Darknet, YOLO, ArcFace, InsightFace, NVIDIA NeMo TitaNet, Resemblyzer

Machine Learning and Deep Learning
Machine Learning, Deep Learning, Neural Networks, PyTorch, TensorFlow, Keras, Gradient Boosting (CatBoost, LightGBM, XGBoost), Recommender Systems, Artificial Intelligence, Scikit-learn

Programming and Tools
Python, Matlab, NetLogo, Git, LaTeX, Flask, RESTful WebServices, Postman, Programming, Pytest, Flask-RESTful, Docker, PyCharm, Jupyter, Anaconda, Linux, Windows

Data
Data Science, Data Analysis, Data Analytics, Data Visualization, Tableau, DASK, SQL, Elasticsearch, NumPy, Pandas, Matplotlib, SQL Server, Faiss

Research and Professional
Research, Research and Development (R&D), Agile Methodologies, Microsoft Office, WordPress

Reviewer

- **Applied Soft Computing** — Elsevier
- **Computer Methods and Programs in Biomedicine** — Elsevier
- **Biomedical Signal Processing and Control** — Elsevier
- **International Journal of Imaging Systems and Technology** — Wiley
- **Cancer Control** — SAGE
- **Journal of Applied Intelligent Systems & Information Sciences (JAISIS)** — FANAP

Licenses and Certifications

- **Pretraining LLMs** — DeepLearning.AI (2025). Credential
- **Finetuning Large Language Models** — DeepLearning.AI (2025). Credential
- **Applied Plotting, Charting and Data Representation in Python** — University of Michigan (Coursera) (2022). Credential
- **Python Project for AI and Application Development** — IBM (2022).
- **Applied Machine Learning in Python** — University of Michigan (Coursera) (2022). Credential
- **Fifth IPM Advanced School on Computing and Artificial Intelligence** — Institute for Research in Fundamental Sciences (IPM) (2021). Credential
- **Fourth IPM Advanced School on Computing and Artificial Intelligence** — Institute for Research in Fundamental Sciences (IPM) (2020). Credential
- **IPM Advanced Data Science Summer School (ADS3)** — Institute for Research in Fundamental Sciences (IPM) (2018).
- **IPM Elementary Data Science Summer School (EDS3)** — Institute for Research in Fundamental Sciences (IPM) (2018).

Teaching Assistant

- **Machine Learning** — Graduate Teaching Assistant, University of Isfahan, Iran (Jan 2016–Jun 2017), under the supervision of Dr. Peyman Adibi.
- **Statistical Pattern Recognition** — Graduate Teaching Assistant, University of Isfahan, Iran (Jan 2016–Jun 2017), under the supervision of Dr. Peyman Adibi.
- **Algorithm Design** — Graduate Teaching Assistant, University of Isfahan, Iran (Jan 2016–Jun 2017), under the supervision of Dr. Peyman Adibi.

Languages

Spoken
Persian (native), English (C1), German (A2)

Interests

Medical Image Analysis
Content-based Image Retrieval, Histopathology, Foundation Models

Graph Neural Networks
Graph Convolutional Networks, Graph Autoencoders, Generative Models

Applied AI
RAG, Speech Pipelines, Face and Speaker Recognition, Recommender Systems

References

Amirreza Mahbod
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Vahid Hajipour
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